

$$\begin{aligned}
 1. \quad & 3x - 2y + 3z = 3 \quad - R_1 \\
 & x + 4y - 9z = 0 \quad - R_2 \\
 & \frac{1}{2}x - y + z = \frac{1}{3} \quad - R_3
 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 3 & -2 & 3 & 3 \\ 1 & 4 & -9 & 0 \\ \frac{1}{2} & -1 & 1 & \frac{1}{3} \end{array} \right]$$

$$R_1 = \frac{R_1}{3}$$

$$\left[\begin{array}{ccc|c} 1 & -2/3 & 1 & 1 \\ 1 & 4 & -9 & 0 \\ \frac{1}{2} & -1 & 1 & \frac{1}{3} \end{array} \right]$$

$$R_2 = R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & -2/3 & 1 & 1 \\ 0 & 14/3 & -10 & -1 \\ \frac{1}{2} & -1 & 1 & \frac{1}{3} \end{array} \right]$$

$$R_3 = R_3 - \frac{R_1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & -2/3 & 1 & 1 \\ 0 & 14/3 & -10 & -1 \\ 0 & -2/3 & \frac{1}{2} & -1/6 \end{array} \right]$$

$$R_2 = \frac{3R_2}{14}$$

$$\left[\begin{array}{ccc|c} 1 & -2/3 & 1 & 1 \\ 0 & 1 & -15/7 & -3/14 \\ 0 & -2/3 & 1/2 & -1/6 \end{array} \right]$$

$$R_1 = R_1 + \frac{2R_2}{3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3/7 & 6/7 \\ 0 & 1 & -15/7 & -3/14 \\ 0 & -2/3 & 1/2 & -1/6 \end{array} \right]$$

$$R_3 = R_3 + \frac{2R_2}{3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3/7 & 6/7 \\ 0 & 1 & -15/7 & -3/14 \\ 0 & 0 & -13/14 & -13/42 \end{array} \right]$$

$$R_3 = -\frac{14}{13} R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3/7 & 6/7 \\ 0 & 1 & -15/7 & -3/14 \\ 0 & 0 & 1 & 1/3 \end{array} \right]$$

$$R_1 = R_1 + \frac{3R_3}{7}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -15/7 & -3/14 \\ 0 & 0 & 1 & 1/3 \end{array} \right]$$

$$R_2 = R_2 + \frac{15R_3}{7}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1/3 \end{array} \right]$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \\ 1/3 \end{bmatrix}$$

$$2x_1 + x_2 + x_3 + x_4 = 8 \quad - R_1$$

$$x_1 - x_2 + x_3 - x_4 = -5 \quad - R_2$$

$$3x_1 - 2x_2 + 3x_3 + x_4 = -1 \quad - R_3$$

$$-x_1 - 4x_2 + 5x_3 - x_4 = -17 \quad - R_4$$

$$\left[\begin{array}{cccc|c} 2 & 1 & -1 & 1 & 8 \\ 1 & -1 & 1 & -1 & -5 \\ 3 & -2 & 3 & 1 & -1 \\ -1 & -4 & 5 & -1 & -17 \end{array} \right]$$

$$R_1 = \frac{R_1}{2}$$

$$\left[\begin{array}{cccc|c} 1 & 1/2 & -1/2 & 1/2 & 4 \\ 1 & -1 & 1 & -1 & -5 \\ 3 & -2 & 3 & 1 & -1 \\ -1 & -4 & 5 & -1 & -17 \end{array} \right]$$

$$R_2 = R_2 - R_1$$

$$\left[\begin{array}{cccc|c} 1 & 1/2 & -1/2 & 1/2 & 4 \\ 0 & -3/2 & 3/2 & -3/2 & -9 \\ 3 & -4 & 3 & 1 & -1 \\ -1 & -4 & 5 & -1 & -17 \end{array} \right]$$

$$R_3 = R_3 - 3R_1$$

$$\left[\begin{array}{cccc|c} 1 & 1/2 & -1/2 & 1/2 & 4 \\ 0 & -3/2 & 3/2 & -3/2 & -9 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \\ -1 & -4 & 5 & -1 & -17 \end{array} \right]$$

$$R_4 = R_4 + R_1$$

$$\left[\begin{array}{cccc|c} 1 & 1/2 & -1/2 & 1/2 & 4 \\ 0 & -3/2 & 3/2 & -3/2 & -9 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \end{array} \right]$$

$$R_2 = -\frac{2R_2}{3}$$

$$\left[\begin{array}{cccc|c} 1 & 1/2 & -1/2 & 1/2 & 4 \\ 0 & 1 & -1 & 1 & 6 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \end{array} \right]$$

$$R_1 = R_1 - \frac{R_2}{2}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & 6 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \end{array} \right]$$

$$R_3 = R_3 + \frac{7R_2}{2}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & 6 \\ 0 & 0 & 1 & 3 & 8 \\ 0 & -7/2 & 9/2 & -1/2 & -13 \end{array} \right]$$

$$R_4 = R_4 + \frac{7R_2}{2}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & 6 \\ 0 & 0 & 1 & 3 & 8 \\ 0 & 0 & 1 & 3 & 8 \end{array} \right]$$

$$R_2 = R_2 + R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 & 14 \\ 0 & 0 & 1 & 3 & 8 \\ 0 & 0 & 1 & 3 & 8 \end{array} \right]$$

$$R_4 = R_4 - R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 & 14 \\ 0 & 0 & 1 & 3 & 8 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 = 1, \quad x_2 + x_4 = 14, \quad x_3 + x_4 = 8$$

$$x_4 = t, \quad x_2 = 14 - x_4, \quad x_3 = 8 - x_4$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 14 - t \\ 8 - t \\ t \end{bmatrix}$$

$$3. \quad x + 3y + 4z = 1 \quad - R_1$$

$$5y - 3z = 26 \quad - R_2$$

$$2x + y = -2 \quad - R_3$$

$$5x + 4y + z = -1 \quad - R_4$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 1 \\ 0 & 5 & -3 & 26 \\ 2 & 1 & 0 & -2 \\ 5 & 4 & 1 & -1 \end{array} \right]$$

$$R_3 = R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 1 \\ 0 & 5 & -3 & 26 \\ 0 & -5 & -8 & -4 \\ 5 & 4 & 1 & -1 \end{array} \right]$$

$$R_4 = R_4 - 5R_1$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 1 \\ 0 & 5 & -3 & 26 \\ 0 & -5 & -8 & -4 \\ 0 & -11 & -19 & -6 \end{array} \right]$$

$$R_2 = \frac{R_2}{5}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 1 \\ 0 & 1 & -3/5 & 26/5 \\ 0 & -5 & -8 & -4 \\ 0 & -11 & -19 & -6 \end{array} \right]$$

$$R_1 = R_1 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 29/5 & -73/5 \\ 0 & 1 & -3/5 & 26/5 \\ 0 & -5 & -8 & -4 \\ 0 & -11 & -19 & -6 \end{array} \right]$$

$$R_3 = R_3 + 5R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 29/5 & -73/5 \\ 0 & 1 & -3/5 & 26/5 \\ 0 & 0 & -11 & 22 \\ 0 & -11 & -19 & -6 \end{array} \right]$$

$$R_4 = R_4 + 11R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 29/5 & -73/5 \\ 0 & 1 & -3/5 & 26/5 \\ 0 & 0 & -11 & 22 \\ 0 & 0 & -128/5 & 256/5 \end{array} \right]$$

$$R_3 = -\frac{R_3}{11}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 29/5 & -73/5 \\ 0 & 1 & -3/5 & 26/5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -128/5 & 256/5 \end{array} \right]$$

$$R_1 = R_1 - \frac{29R_3}{5}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & -3/5 & 26/5 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -128/5 & 256/5 \end{array} \right]$$

$$R_2 = R_2 + \frac{3R_3}{5}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -128/5 & -256/5 \end{array} \right]$$

$$R_4 = R_4 + \frac{128R_5}{5}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \\ -2 \end{bmatrix}$$

$$4. \quad 3x_1 + 2x_2 + 4x_3 = 8$$

$$x_2 - x_3 + 2x_4 + 5x_6 = -2$$

$$-5x_1 + 2x_2 + 3x_3 - 9x_6 = 9$$

$$7x_1 - 3x_4 + x_5 = 2$$

$$\left[\begin{array}{cccccc|c} 3 & 2 & 4 & 0 & 0 & 0 & 8 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ -5 & 2 & 3 & 0 & 0 & -9 & 9 \\ 7 & 0 & 0 & -3 & 1 & 0 & 2 \end{array} \right]$$

$$R_1 = \frac{R_1}{3}$$

$$\left[\begin{array}{cccccc|c} 1 & 2/3 & 4/3 & 0 & 0 & 0 & 8/3 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ -5 & 2 & 3 & 0 & 0 & -9 & 9 \\ 7 & 0 & 0 & -3 & 1 & 0 & 2 \end{array} \right]$$

$$R_3 = R_3 + 5R_1$$

$$\left[\begin{array}{cccccc|c} 1 & 2/3 & 4/3 & 0 & 0 & 0 & 8/3 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ 0 & 16/3 & 29/3 & 0 & 0 & -9 & 67/3 \\ 7 & 0 & 0 & -3 & 1 & 0 & 2 \end{array} \right]$$

$$R_4 = R_4 - 7R_1$$

$$\left[\begin{array}{cccccc|c} 1 & 2/3 & 4/3 & 0 & 0 & 0 & 8/3 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ 0 & 14/3 & 29/3 & 0 & 0 & -9 & 67/3 \\ 0 & -14/3 & -28/3 & -3 & 1 & 0 & -50/3 \end{array} \right]$$

$$R_1 = R_1 - \frac{2R_2}{3}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 2 & -4/3 & 0 & -10/3 & 4 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ 0 & 16/3 & 29/3 & 0 & 0 & -9 & 67/3 \\ 0 & -14/3 & -28/3 & -3 & 1 & 0 & -50/3 \end{array} \right]$$

$$R_3 = R_3 - \frac{16R_2}{3}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 2 & -4/3 & 0 & -10/3 & 4 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ 0 & 0 & 15 & -32/3 & 0 & -107/3 & 33 \\ 0 & -14/3 & -28/3 & -3 & 1 & 0 & -50/3 \end{array} \right]$$

$$R_4 = R_4 + \frac{14R_2}{3}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 2 & -4/3 & 0 & -10/3 & 4 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ 0 & 0 & 15 & -32/3 & 0 & -107/3 & 33 \\ 0 & 0 & 14 & 19/3 & 1 & 70/3 & -26 \end{array} \right]$$

$$R_3 = \frac{R_3}{15}$$

$$\left[\begin{array}{cccccc|c} 1 & 0 & 2 & -4/3 & 0 & -10/3 & 4 \\ 0 & 1 & -1 & 2 & 0 & 5 & -2 \\ 0 & 0 & 1 & -32/45 & 0 & -107/45 & 11/5 \\ 0 & 0 & -14 & 19/3 & 1 & 70/3 & -26 \end{array} \right]$$

$$R_1 = R_1 - 2R_3$$

$$\left[\begin{array}{ccc|cc|c} 1 & 0 & 0 & 41/45 & 0 & 64/45 & -2/5 \\ 0 & 1 & -1 & \cancel{2/45} & 0 & 5 & -2 \\ 0 & 0 & 1 & -32/45 & 0 & -107/45 & 11/5 \\ 0 & 0 & -14 & 19/3 & 1 & 70/3 & -26 \end{array} \right]$$

$$R_2 = R_2 + R_3$$

$$\left[\begin{array}{ccc|cc|c} 1 & 0 & 0 & 41/45 & 0 & 64/45 & -2/5 \\ 0 & 1 & 0 & 58/45 & 0 & 118/45 & -1/5 \\ 0 & 0 & 1 & -32/45 & 0 & -107/45 & 11/5 \\ 0 & 0 & -14 & 19/3 & 1 & 70/3 & -26 \end{array} \right]$$

$$R_4 = R_4 + 14R_3$$

$$\left[\begin{array}{ccc|cc|c} 1 & 0 & 0 & 41/45 & 0 & 64/45 & -2/5 \\ 0 & 1 & 0 & 58/45 & 0 & 118/45 & 1/5 \\ 0 & 0 & 1 & -32/45 & 0 & -107/45 & 11/5 \\ 0 & 0 & 0 & -163/45 & 1 & -448/45 & 24/5 \end{array} \right]$$

$$R_4 = -\frac{45R_4}{163}$$

$$\left[\begin{array}{ccc|cc|c} 1 & 0 & 0 & 41/45 & 0 & 64/45 & -2/5 \\ 0 & 1 & 0 & 58/45 & 0 & 118/45 & 1/5 \\ 0 & 0 & 1 & -32/45 & 0 & -107/45 & 11/5 \\ 0 & 0 & 0 & 1 & -45/163 & 448/163 & -214/163 \end{array} \right]$$

$$R_1 = R_1 - \frac{4R_4}{45}$$

$$\left[\begin{array}{cccc|cc|c} 1 & 0 & 0 & 0 & 4/163 & 192/163 & -46/163 \\ 0 & 1 & 0 & 58/45 & 0 & 118/45 & 1/5 \\ 0 & 0 & 1 & -32/45 & 0 & -107/45 & 11/5 \\ 0 & 0 & 0 & 1 & -45/163 & 448/163 & -216/163 \end{array} \right]$$

$$R_2 = R_2 - \frac{58R_4}{45}$$

$$\left[\begin{array}{cccc|cc|c} 1 & 0 & 0 & 0 & 4/163 & 192/163 & -46/163 \\ 0 & 1 & 0 & 0 & 58/163 & -150/163 & 311/163 \\ 0 & 0 & 1 & -32/45 & 0 & -107/45 & 11/5 \\ 0 & 0 & 0 & 0 & -45/163 & 448/163 & -216/163 \end{array} \right]$$

$$R_3 = R_3 + \frac{32R_4}{45}$$

$$\left[\begin{array}{cccc|cc|c} 1 & 0 & 0 & 0 & 4/163 & 192/163 & -46/163 \\ 0 & 1 & 0 & 0 & 58/163 & -150/163 & 311/163 \\ 0 & 0 & 1 & 0 & 0 & -107/45 & 11/5 \\ 0 & 0 & 0 & 1 & -45/163 & 448/163 & -216/163 \end{array} \right]$$